

PatientTriage.ai: Business Proposal & Enterprise Strategy

“Triage is a snapshot. Risk isn't.” — Closing the Waiting-Room Surveillance Gap

AI recommends. Deterministic safety rules protect. Clinicians decide.

Accenture Innovation Challenge 2026 • Round 2 Business Case • Modeled 500-Bed ED Impact

GROSS ANNUAL VALUE \$3.82M / Hospital	ESTIMATED NET ROI \$3.58M / yr (14.9x)	LWBS REVENUE RECOVERY +\$1.12M / yr (30%)	TARGET MARKET 5,500+ US/EU EDs
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Executive Summary

Triage determines priority at the front door—but patients don't remain static while they wait. With over 140M annual US visits and acute nurse turnover (26.8%), ESI Level 3/4 patients wait 2.5 to 4.5 hours unmonitored; silent decompensations (hypoxia, sepsis, shock) frequently remain undetected until waiting-room collapse. PatientTriage.ai is an AI Safety Copilot purpose-built for the emergency waiting room. It combines continuous vital trajectory surveillance, observation shelf-life decay, uncertainty scoring (*Unknown ≠ Safe*), and clinician attention discounting (The Attention Gap) to generate **\$3.82M in gross annual value (\$3.58M net ROI)** per 500-bed facility.

1. Market Opportunity & Problem Framing (The Surveillance Blind Spot)

- **Silent Waiting Room Deterioration:** Patients classified as stable at intake deteriorate silently while waiting; 79% of ED staff report triage bottlenecks as their primary clinical risk.
- **Missing & Stale Data Handling:** When critical observations are missing or stale, conventional workflows may lack sufficient evidence to safely reassess waiting patients. PatientTriage.ai treats missing or unreliable data as uncertainty rather than reassurance.
- **Competitive Moat vs. Epic & Cerner:** Inpatient algorithms score static bedded severity. PatientTriage.ai is purpose-built for the waiting room and accounts for physician coverage and evidence staleness decay.

2. Continuous Patient Journey & 4-State Clinical Dispatch

Patient Arrives ■■■ Triage Baseline ■■■ Waiting Lounge (2.5–4.5h) ■■■ Trajectory Check ■■■ 4-State Queue (■/■/■/■) ■■■ Reassess ■■■ Escalate

State	Physiological Definition	Clinical Action Triggered
■ CONTINUE	Vitals stable within safe limits; observation shelf-life active.	Maintain waiting room monitoring.
■ REASSESS	Observation shelf-life expired (stale data) or moderate vital drift.	Dispatches nurse bedside recheck round.
■ ESCALATE	Deterministic red-flag breach (SpO ₂ ■ < 85%, SBP < 75 mmHg).	Immediate resuscitation bay allocation & MD page.
■ UNCERTAIN	Missing intake vitals, sensor disconnect, or conflicting metrics.	Forces physical human verification (<i>Unknown ≠ Safe</i>).

3. Target Stakeholders & Enterprise Value Matrix

Stakeholder	Primary Clinical / Financial Pain Point	PatientTriage.ai Value Delivery
Triage Nurses (RNs)	Overwhelmed tracking 40+ waiting patients; alarm fatigue.	Ranked 'Next 5 Mins' queue surfaces top 3 actionable tasks.
Emergency MDs	Blind to which waiting patient worsened since intake.	Attention Gap Queue dispatches clinicians to highest unmet need.
Chief Medical Officers	Delayed diagnosis lawsuits and sentinel waiting-room events.	Deterministic guardrails & append-only audit trail for malpractice defense.
Hospital CFOs	Uncompensated ICU crashes, LWBS revenue leakage, turnover.	Delivers modeled \$3.58M net annual ROI (14.9x return on license).

4. Business Case, Financial ROI & Impact Model (500-Bed Hospital Facility)

Modeled Financial & Clinical Projections based on 65,000 annual ED visits, 500 acute beds, and \$1,200 average ED revenue/visit (Derived from published emergency health economics literature; requires hospital-specific validation):

Value Driver	Pre-Implementation Baseline	Modeled Post-Implementation Impact	Annual Financial Value
1. LWBS Revenue Recovery	3,120 patients/yr leave (4.8%)	30% reduction via proactive re-engagement	+\$1,123,200 / yr
2. Avoided ICU Transfers	145 waiting room ICU crashes/yr	64% reduction (93 avoided ICU stays @ \$15k)	+\$1,395,000 / yr
3. Malpractice Risk Mitigation	\$1.2M annual liability reserve	40% reduction via documented audit ledger	+\$480,000 / yr
4. Nurse Retention & Overtime	26.8% nurse turnover (14 replacements)	4 replacements avoided + 15% overtime reduction	+\$378,000 / yr
5. ED Throughput & Boarding	248 mins average wait/boarding	30-minute reduction via optimized dispatch	+\$445,000 / yr
TOTAL GROSS ANNUAL VALUE	—	—	\$3,821,200 / yr
Enterprise License & Support	—	Annual Software Subscription & Edge Hardware	-\$240,000 / yr
ESTIMATED NET ANNUAL ROI	—	14.9x Net Return on Investment	+\$3,581,200 / yr

5. Technical Interoperability & HL7 FHIR Integration

EHR Interoperability: Ingests HL7 FHIR standard resources (Encounter, Observation) via SMART-on-FHIR and CDS Hooks (patient-view). Seamlessly overlays existing Epic and Cerner installations without disrupting core clinical workflows.

Edge Architecture: Runs air-gapped on local hospital servers with sub-15ms inference latency, zero external cloud LLM dependencies, and zero persistent PHI in the ranking cache.

6. Phased Enterprise Roadmap: Core Deployment to Regional Extensions

- **Phase 1 (Q3 2026 - Completed):** Lab Benchmark Validation • 20 synthetic cohorts verified across 51 automated unit/integration tests; sub-15ms latency (*validates prototype algorithmic behavior only; not clinical efficacy proof*).
- **Phase 2 (Q4 2026):** Shadow Clinical Trial • Silent background deployment alongside Epic/Cerner via HL7 FHIR; clinician concordance evaluation.
- **Phase 3 (Q1–Q2 2027):** Live Hospital Pilot • Primary Safety Endpoint: >45% reduction in Mean Time to Escalation (MTTE); Operational Endpoint: <2 non-actionable alerts/nurse/shift; Economic Endpoint: ≥25% LWBS reduction.
- **Phase 4 (Q3 2027+):** Multi-Hospital Extensions • Regional network load balancing, 108 EMS ambulance pre-arrival telemetry, and referral diversion.

7. Risk Management, Fail-Safe Behavior & Regulatory Compliance

- **Deterministic Red-Flag Override:** Hard physiological red-flags override all statistical scores; downgrade safety blocker prevents unsafe de-escalations.
- **Clinician Alarm Fatigue Mitigation:** Queue compression surfaces only the top actionable tasks in the 'Next 5 Mins' queue, preventing alarm flooding.
- **Fail-Safe Protocol:** If network connectivity is lost, the system fails safe to manual clinical rounding with deterministic guardrails active.
- **Regulatory Stance:** Positioned as Clinical Decision Support under FDA Non-Device CDS (21 U.S.C. § 360aaa-1); licensed clinicians retain 100% final override authority.